Elliptic Curves & Cryptography

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Books Used

Presentation2/Screenshot 2026-04-15

Figure: An Introduction To Mathematical Cryptography

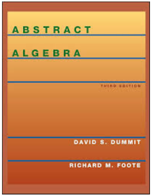

Figure: Dummit And Foote's Abstract Algebra

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Public Key Cryptography

The Discrete Log Problem

Let g be an element of a group G. **The Discrete Log Problem** for G is the following:

Definition

Discrete Log Problem: Given some element h in the subgroup generated by g, find the smallest integer $m > 0$ which satisfies $h = g^m$.

The smallest m to satisfy $h = g^m$ is known as the discrete log of h with respect to g.

Hardness Of The Discrete Log Problem

How hard is the Discrete Log Problem? It depends upon the group!
For some groups the discrete log problem is more difficult than others. For example:

$$\mathbb{F}_p^* \text{ (elements } \mathbb{F}_p \text{ under multiplication)}$$

come back to slide... The best known algorithm to solve this DLP is of time:

$$O(e^{\sqrt{(\log p)(\log \log p)^2}})$$

Which is known as **subexponential** time. (Faster than exponential but slower than polynomial.)

Diffie-Hellman Key Exchange

The **Diffie-Hellman Key Exchange** allows two parties to exchange a secret key over an insecure public network.
Public Knowledge: Some group G and some $g \in G$ or order n.

Definition

- Bob and Alice choose some secret $0 < b < n$ and $0 < a < n$ respectively.
- Each computes $B = g^b$ and $A = g^a$ and sends those values to each other.
- Each computes A^b and B^a Bob and Alice now have the shared value:
$$A^b = g^{ab} = g^{ba} = B^a$$

Computing the value g^{ab} depends on the difficulty of solving the discrete log problem for g^a and g^b.

Diffie-Hellmen Key Exchange Example

Let us introduce an example of Diffie Hellmen Key Exchange over $\mathbb{Z}/941\mathbb{Z}$:

- Alice and Bob agree to some $p = 941$ and primitive root $g = 627$
- Alice chooses secret $a = 347$ and computes: $A = 390 \equiv 627^{347} \pmod{941}$
- Similarly Bob chooses secret $b = 781$ and computes: $B = 691 \equiv 627^{781} \pmod{941}$
- Finally, they have shared value:

$$470 \equiv 627^{347 \cdot 781} \equiv A^b \equiv B^a \pmod{941}$$

Why Elliptic Curves?

- Elliptic curves with points in $\mathbb{F}_p$ are finite groups. (We can translate traditional cryptography problems to elliptic curves over finite fields)
- This group is "general enough" so there is no especially efficient algorithm to compute its associated Discrete Log Problem.
- Elliptic curves are used in cryptography because of their associated "hardness" for solving problems.
- Translations of classical cryptography problems such as the Discrete Log Problem to elliptic curves allows for an increase in hardness, smaller key sizes, etc.

Definition: Elliptic Curves

Definition

Formally, an **Elliptic Curve** over some field K, is a smooth projective curve of genus 1 with at least one K-Rational point (There is at least one point, P, on E, with coordinates in K).

We define the set of **K-Rational Points** on an Elliptic Curve E as the set of points on E whose coordinates all lie in K as well as the point $\mathcal{O}$. Denoted by $E(K)$.

Weierstrass Equation

We can define an **Elliptic Curve** with the following **Weierstass Equation:**

$$Y^2 = X^3 + BX + C$$

with a discriminant of:

$$\Delta = 4A^3 + 27B^2 \neq 0$$

The Weierstrass Equation has no repeated roots, making this curve non-singular. Additionally an elliptic curve has a point at infinity which we will denote as $\mathcal{O}$. We define E to be the set:

$$E = \{(x, y) : y^2 = x^3 + bx + c\} \cup \{\mathcal{O}\}$$

Finite Fields

In the context of cryptography we are particularly interested in elliptic curves over finite fields, namely $\mathbb{F}_p$.

$$E : Y^2 = X^3 + AX + B$$

$$A, B \in \mathbb{F}_p \text{ and } 4A^3 + 27B^2 \neq 0$$

Where points on this curve are:

$$E(\mathbb{F}_p) = \{(x, y) : x, y \in \mathbb{F}_p \text{ and } y^2 = x^3 + bx + c\} \cup \{\mathcal{O}\}$$

Additionally, I want to put emphasis on the fact that we will be dealing with elliptic curves over $\mathbb{R}$ from this point forward.

Weierstrass Equation

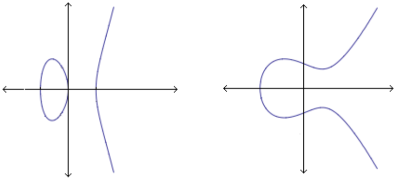

Figure: Typical Shapes Of Elliptic Curves of form $y^2 = x^3 + ax + b$

Complex Numbers

An Amazing Fact:

An Elliptic curve over the Complex Numbers, $\mathbb{C}$, is a manifold in the form of $\mathbb{C}/\mathbb{L}$, ($\mathbb{C}$ modulo a Lattice), where $\mathbb{L}$ is a lattice in the complex plane.

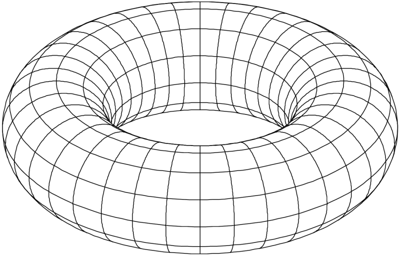

Figure: Torus

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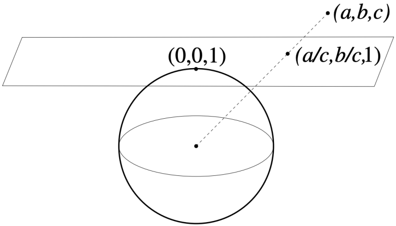
The Points At Infinity

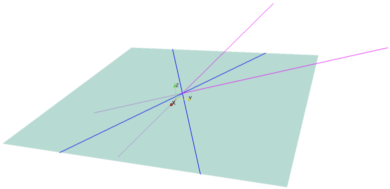
Just for a little bit of intuition we will discuss what I mean by "the point at infinity".

In projective space points (in the form of $(x : y : z)$) are represented as lines through the origin of $\mathbb{R}^3$. We can imagine an Affine Plane to be embedded in this projective space on $z = 1$.



The Points At Infinity

Projective points that lie on $z = 0$ (in the form of $(x : y : 0)$) are points at infinity. Imagine living on the Affine Plane, those points parallel to the Affine Plane (points at infinity) will be impossible to reach.



- Blue: Points At Infinity
- Pink: Affine Points

Addition

We are interested in adding two points on a curve to get a third point:

Addition Of Points

Let P and Q be two points on E. We draw a line L through P and Q. This Line intersects E through at most three points. The third point we denote as R. We then take R and reflect it across the x-axis and get R' , where R' is the sum of P + Q. This "addition" operation we denote with the symbol $\oplus$.

Addition

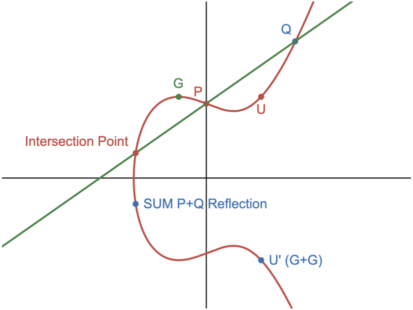


Figure: The sum of $P \oplus Q$ reflected across x-axis

Adding Some Point To Itself

Taking a point G and adding it to itself? We observe that $G \oplus G = 2G = U'$ results in:

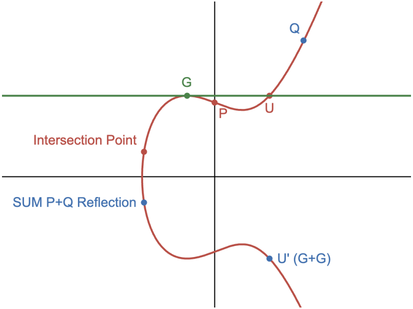


Figure: A tangent line to point G through point U reflected across x-axis to yield U' which is $2G$.

In this case the slope of the tangent line can be found by implicitly differentiating the Weierstrass Equation.

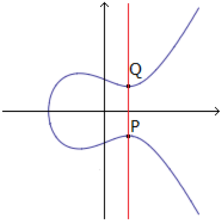
Intersection At Infinity

Previously we mentioned the point at infinity. We "extend a line to the point at infinity".
Taking a look at $P = (a, b)$ and it's x-axis reflection $P' = (a, b)$
The line L would be a vertical line through P, P'.

$$P \oplus P' = \mathcal{O}$$

L intersects the point at infinity. Additionally:

$$P + \mathcal{O} = P$$



Addition Formula

We can describe a formula for adding points on E using elementary methods:

Addition Formula

- If $P_1 = P_2$ and $x_1 = x_2$ then $P_1 + P_2 = \mathcal{O}$
- If $P_1 = P_2$ and $y_1 = 0$ then $P_1 + P_2 = 2P_1 = \mathcal{O}$
- If $P_1 \neq P_2$ and $x_1 \neq x_2$ then let $\lambda = \frac{y_2 - y_1}{x_2 - x_1}$ and $v = \frac{y_1 x_2 - y_2 x_1}{x_2 - x_1}$
- If $P_1 = P_2$ and $y_1 \neq 0$ then let $\lambda = \frac{3x_1^2 + A}{2y_1}$ and $v = \frac{-x^3 + Ax + 2B}{2y}$

Then:

$$P_1 + P_2 = (\lambda^2 - x_1 - x_2, -\lambda^3 + \lambda(x_1 + x_2) - v)$$

Frame Title

We essentially just showed that the addition on elliptic eurves makes an abelian group with $\mathcal{O}$ as the identity element.

Theorem

The addition law on E has the following properties:

- *Identity Element:* $P + \mathcal{O} = \mathcal{O} + P$
- *Inverse:* $P + (-P) = \mathcal{O}$
- *Commutative:* $P + Q = Q + P$
- *Associative:* $P + (Q + R) = (P + Q) + R$

While it is quite easy to see (and prove) the first three properties, it is not that intuitive, and quite arduous to prove associativity.

Elliptic Curves Over Finite Fields Example

Consider The Elliptic Curve:

$$E : X^3 + 3X + 8, \text{ \emph{over the field} } \mathbb{F}_{13}$$

We can find points on $E(\mathbb{F}_{13})$ by subbing in points $X = \{1, 2, 3, ..., 12\}$ and checking to see if X is a square $\pmod{13}$. Take $X = 1$, we find that $(1)^3 + 3(1) + 8 \equiv 12 \pmod{13}$, where

$$5^2 \equiv 12 \pmod{13} \text{ and } 8^2 \equiv 12 \pmod{13}$$

We now have points $(1, 5), (1, 8)$.

$E(\mathbb{F}_{13}) = \{\mathcal{O}, (1, 5), (1, 8), (2, 3), (2, 10), (9, 6), (9, 7), (12, 2), (12, 11)\}$

E($\mathbb{F}_p$) as a group

It is clear that the set of points in $E(\mathbb{F}_p)$ is finite (Finite number of possibilities for X, Y). *How many elements (points) are inside some arbitrary $E(\mathbb{F}_p)$?*

Theorem

Hasse' Theorem: Let E be an Elliptic Curve over $\mathbb{F}_p$ The number of elements in $E(\mathbb{F}_p)$ (denoted by $\#E(\mathbb{F}_p)$) is:

$$\#E(\mathbb{F}_p) = p + 1 - t_p$$

$$|t_p| \leq 2\sqrt{p}$$

Where t_p is the **Trace of Frobenius**.

Elliptic Curve Discrete Log Problem

Definition

The **Elliptic Curve Discrete Log Problem (ECDLP)** goes as follows: Let E be an elliptic curve defined over $\mathbb{F}_p$

$$E : y^2 = x^3 + Ax + B \text{ and } A, B \in \mathbb{F}_p$$

Take points $P, Q \in E(\mathbb{F}_p)$. We must find some $n \in \mathbb{Z}$ such that

$$Q = nP$$

We denote the value n as the **Discrete Logarithm** of Q:

$$n = \log_P(Q)$$

Solving ECDLP

"Brute Force" Method

Compute the values n_1P, n_2P, n_3P until finding that value to satisfy $nP = Q$. The expected running time here is $O(p)$, since $\#E(\mathbb{F}_p) = O(p)$. Not very efficient.

Collision Search

Compute two lists for randomly chosen values $n_1, n_2, n_3, \ldots$

$$\text{List 1: } n_1P, n_2P, n_3P\ldots$$

$$\text{List 2: } Q - n_1P, Q - n_2P, Q - n_3P\ldots$$

until we find a "collision":

$$n_iP = Q - n_jP$$

Running time is $O(\sqrt{p})$ (Which is fastest known time to solve).

So Why Elliptic Curves?

- We actually find that $O(\sqrt{p})$ is the fastest known method to solve ECDLP. This means that it is not feasible to solve ECDLP in $E(\mathbb{F}_q)$ if $q > 2^{160}$.
- A standard DLP equivalent difficulty in (say) $\mathbb{F}_q^*$ requires that $q \approx 2^{1000}$.
- ECDLP with $q \approx 2^{160}$ is about as hard as factoring a 1000 bit number.
- *Hence we find that an elliptic curve based cryptography system requires a smaller key, is more efficient, and is potentially faster.*

Elliptic Curve Diffie-Hellman Key Exchange

Method:

Two Allies Alice and Bob agree to some $E(\mathbb{F}_p)$ and some point $P \in E(\mathbb{F}_p)$, each choose some secret n_A, n_B respectively, where they compute:

$$Q_A = n_A P \text{ and } Q_B = n_B P$$

Upon exchanging Q_A and Q_B, they use their respective secret values n_A, n_B to compute:

$$n_A Q_B = n_B Q_A = n_A n_B P$$

Where $n_A n_B P$ is their shared secret value.

Elliptic Curve Diffie-Hellman Key Exchange (Example)

$$ATOX\ \ T\ \ ZHHW\ \ LNFFXK \text{ (Decrypt me!)}$$

Take the following elliptic curve: $p = 3851$, $E : Y^2 = X^3 + 324X + 1287$ with the point $P(920, 303) \in E(\mathbb{F}_{3851})$.

Computations:

- Alice and Bob choose their secret Values: $n_a = 1194$, $n_b = 1759$.
- Alice and Bob each compute:
  $Q_A = 1194(P) = (2067, 2178) \in E(\mathbb{F}_{3851})$
  $Q_B = 1759(P) = (3684, 3125) \in E(\mathbb{F}_{3851})$
- After exchanging these values, Alice and Bob Compute:
  $n_a Q_B = 1194(3684, 3125) = (3347, 1242) \in E(\mathbb{F}_{3851})$
  $n_a Q_A = 1759(2067, 2178) = (3347, 1242) \in E(\mathbb{F}_{3851})$

Elliptic Curve Diffie-Hellman Key Exchange (Example Continued)

We found that Alice and Bob share the point:

$$(3347, 1242) \in E(\mathbb{F}_{3851})$$

Where they will take the x coordinate to be their secret key. A basic **shift cipher** uses the formula:

Shift Cipher

$$c \equiv (m + K) \pmod{26} \implies m \equiv (c - K) \pmod{26}$$

(c - cipher text, m -plaintext, K - secret key)

Using 3347 as our secret key we find the message to be...

Elliptic Curve Diffie-Hellman Key Exchange (Example Continued)

Figure: Have A Good Summer!

Acknowledgments

write later.